

Liberal Studies Culminating Project 2012



Convolution Reverb Impulse Response Library

By Greg Hopkins
With Faculty Mentor, Professor John Hill

Project Overview

The goal of this project is to create a library of no less than 10 impulse responses which may be used for applying artificial reverb that models actual physical spaces and hardware reverb devices. These spaces are on the campus of Middle Tennessee State University and the hardware reverb devices are a combination of school property, personal property, and rental equipment. Useful in music and film production, this library is made available under a Creative Commons license for non-commercial use. In addition, a documentation video is provided which demonstrates the techniques used to capture the physical spaces, and demonstrates the use of the resulting impulse response. Of particular interest is the intent to demonstrate the ability to create useful impulse responses with what an engineer might reasonably have on hand. As this project shows, it is not necessary to purchase high end specialty equipment in order to produce reasonable results. While the same techniques are used by acousticians to quantitatively measure the acoustics of a space, it is not the intent of this project to achieve accuracy, so much as useability for the reproduction of unique reverbs.

Methods Used

After a period of trial and error, two methods for the creation of the impulse responses have been chosen for this project as described by Ben-Hador and Neoran.¹ These methods are paraphrased as follows:

Burst Method – A short burst of sound is generated to excite the space, which is recorded by a microphone array and is ready for use with only minimal post processing. Methods for generating the burst of sound vary widely, and will have a direct impact on the timbre of the resulting reverb. The generator should produce a wide frequency spectrum burst to achieve full bandwidth reverb. In determining the generating method, several devices have been evaluated for their ability to excite a space across a wide frequency band. A starter pistol, slapstick, balloon pop, or fire cracker will generate a burst with sufficient amplitude to excite the space. However, using a starter pistol or fire cracker is not a viable option for indoor use or where city and state ordinances prohibit their use. From the trials conducted during the course of this project, a slapstick was found to produce too few low frequencies for a balanced reverb, and a balloon pop was found to create too few high frequencies. These methods skewed the reverb's tone toward one end of the frequency spectrum or the other. Nevertheless, a selection of spaces were captured with the use of a slapstick, and have been included in the library to demonstrate this method. While the results are inferior to those of other methods, the ease of capture is superior and might be of use for some creative purposes. These spaces are filed under [Real Spaces > Slapstick].

Sine Sweep Method – A sine wave sweep is generated which covers the entire audible spectrum over a

¹ Ben-Hador, R., Neoran, I. (2001). Capturing Manipulation and Reproduction of Sampled Acoustic Impulse Responses.

chosen period of time and is reproduced with a loudspeaker to excite the space. The period used in this project was 15 seconds, and the sweep covers 40Hz to 24kHz. According to Lee and Carera, this method has the advantage of averaging out the inherent noise of a space caused by HVAC and fluorescent lighting, thus providing a lower noise floor and more accurate results.² While this project is focused more on musicality than scientific accuracy, the sine sweep method more closely reproduces the acoustic properties of the actual space than did the burst method.

Equipment Used For Sine Sweep Generation And Capture

Apple iPad, iPhone
JBL LSR2328 Active Studio Monitor
Zoom H4n Portable Recorder
AKG C414 XLII Microphones
Shure SM80 Microphones
Pro Tools 10HD

Software Used For Audio Processing

Audacity 2.02
Pro Tools 10HD
Voxengo Deconvolver
Waves IR-1 Convolution Reverb Plugin

Library Description

The library is organized into three collections:

- Analog Reverb Devices
 - EMT 140 Analog Plate Reverb
 - Ecoplate Analog Plate Reverb
 - AKG BX20E1 Analog Spring Reverb
- Digital Reverb Devices
 - Lexicon 480L
 - Lexicon PCM60
 - Lexicon 300
 - Yamaha Rev7
- Real Spaces
 - MTSU Hinton Hall
 - Murfree Center – Memorial Gym Tunnel
 - MTSU Student Union Stairwell
 - MTSU Studio B Live Room
 - Slapstick
 - Bragg Mass Communications Building
 - Studio A Lab Hallway
 - Emergency Stairwell
 - Murfree Center – Memorial Gym Tunnel

Other physical spaces were captured, with the use of both methods, but they were deemed unsuitable to

² Lee, D., Cabrera, D. (2010). Effect of listening level and background noise on the subjective decay rate of room impulse responses: Using time-varying loudness to model reverberance.

include in this library due to high ambient noise levels or simply uninteresting aesthetics. Regardless of these exclusions, the three collections total 223 unique impulse responses.

Deviations From The Project Proposal

Perhaps the greatest deviation from the initial proposal was the inclusion of hardware reverb devices and the division of the library into three collections. The hardware devices chosen were exhaustively captured to provide a wide array of programs, and represent hardware considered to be “classic” studio reverb devices. For many engineers, these devices are now unavailable due to their age or are prohibitively expensive to own. These impulses are therefore more valuable than the less than stellar acoustic spaces omitted.

Initially, the sine sweep method was not going to be used because of its greater difficulty to implement. More equipment is necessary with this method, and a great deal more post processing is required. However, experimentation with alternate methods led to the decision to use sweeps exclusively. It is the estimation of the technician that in total, approximately 20 minutes per impulse response was required to capture, process, and edit each file for use. While this approach required more effort, the improvement in quality was substantial enough to warrant the additional time necessary to employ this method. While the estimated time per impulse seems trivial, in the aggregate, the included content alone represents nearly 75 hours of work. This does not include the trial period or the creation of the documentary video.

The original proposal mentioned the use of two different microphone arrays to provide a comparison between different capture techniques. Changes to the method were explored mid-project, however all included acoustic impulses were captured using either ORTF or AB spaced pair microphone arrays as originally described.

The proposal detailed “true stereo” impulse responses which allow the engineer to pan the dry source audio to specific places in the virtual room. After further research and inspection of many of the spaces and hardware available, the geometry of the space or the programs of the hardware did not always warrant or allow the ability to pan left or right. In the case of a tunnel or hallway, the space is too narrow and reverberant to provide clear directional cues from discrete reflections as described by Deruty³, and many of the hardware reverb devices have only one input for its two outputs. Where appropriate, a mono input to stereo output approach was employed, with alternate microphone or speaker positions to add variety and highlight diffuse reflections. The alternate positions or programs may be used in a front/rear configuration if desired, but asymmetry of the space or drastic differences in programs can potentially cause the reverb to sound off balance and should mostly only be used for special effect.

Challenges Encountered

As expected, tracking down the persons needed to provide after hours access to many of these spaces was difficult. Many phone calls and emails went unanswered. Most spaces listed for potential capture were unsuitable because of high ambient noise or were simply uninteresting and did not meet the technician's standards for this library. The high powered lights in Bragg Mass comm emit a pronounced

3 Deruty, E. (2010, September). Creative Convolution: New Sounds From Impulse Responses.

buzz, and fluorescent lights in many of the spaces do not operate silently. In the Business and Aerospace building, the lack of accessible power outlets in smaller lounges and the high amount of noise from the concessions refrigerators in the main entrance made these spaces impossible to capture with reasonable success. This theme was repeated in most locations visited. The Real Spaces collection represents the best results among the spaces visited.

Timeline

Adhering to the rough timeline provided in the project proposal was nearly impossible. This project was a learning and research expedition, and some changes in workflow and methods were to be expected.

Instructional Documentary Video Production

The instruction video documents the capture of impulse responses in the MTSU Student Union Building. In lieu of a PDF guide for the application of the responses, the video includes a step by step demonstration of the edit and application process, including screenshots of the steps necessary to deploy convolution reverb plugins to make use of the impulse responses. This video shows the use of the Waves IR convolution reverb plugin mentioned by Walker.⁴ The video was shot with two cameras, and edited in Final Cut Pro. Narration was recorded with Pro Tools 10. The video required approximated 10 hours to shoot, edit, narrate, and render.

Review of Results

It is the opinion of the technician that the library as a whole represents a high level of quality and useability. The combination of real spaces and hardware devices adds considerable value and sufficiently provides a diverse selection of reverberant sounds. For those who wish to embark on a similar endeavor to capture and use custom impulse responses, it is recommended that the engineer first visit potential spaces to measure the noise floor with a sound pressure meter before attempting to create the impulse responses. Based on the experience gained from this project, a noise floor of -60dBSPL or lower is recommended for suitable results. Ambient noise above this level requires unnatural fades be applied, which result in awkward sounding reverb tails.

In retrospect, the proximity to Nashville's rich recording industry community could have allowed this project to focus *exclusively* on the capture of esoteric devices by partnering with area equipment rental companies to source rare equipment. Many local professional recording studios have antique analog reverb devices like the EMT 140 plate reverb or AKG BX20 Spring reverb that are included in this project. Since no two analog reverbs of the same model sound exactly alike, including a variety of plate reverbs would make for a truly unique library. Alternatively, some of the landmark Nashville studios have iconic recording spaces and echo chambers which can be heard on countless hit productions. The collection of impulse responses from these spaces would require a great deal more time and coordination to accomplish, but could potentially result in a library which rivals some of the commercially available collections described by Martin Walker. If this idea is realized, a more accurate approach and higher quality equipment configuration is recommended to match the acoustic properties of these spaces as closely as possible.

⁴ Walker, M. (2005, April). Convolution Processing with Impulse Responses.

References

- Ben-Hador, R., Neoran, I. (2001). Capturing Manipulation and Reproduction of Sampled Acoustic Impulse Responses. *Audio Engineering Society Convention Papers*, 117.
- Lee, D., Cabrera, D. (2010). Effect of listening level and background noise on the subjective decay rate of room impulse responses: Using time-varying loudness to model reverberance. *Applied Acoustics*, 71(9), 801-811. doi:0003-682X.
- Deruty, E. (2010, September). Creative Convolution: New Sounds From Impulse Responses. *Sound On Sound | Recording Techniques | Audio Technology | Music Production | Computer Music | Video Media*. Retrieved September 9, 2012, from <http://www.soundonsound.com/sos/sep10/articles/convolution.htm>
- Walker, M. (2005, April). Convolution Processing with Impulse Responses. *Sound On Sound*. Retrieved September 9, 2012, from <http://www.soundonsound.com/sos/apr05/articles/impulse.htm>